

§ 11.6 The Ratio Test(비판정법)

Ex.3 까지

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The Ratio Test

(i) If $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = L < 1$, then the series $\sum_{n=1}^{\infty} a_n$ is absolutely convergent
(and therefore convergent).

(ii) If $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = L > 1$ or $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \infty$, then the series $\sum_{n=1}^{\infty} a_n$ is
divergent.

(iii) If $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = 1$, the Ratio Test is inconclusive; that is, no conclusion
can be drawn about the convergence or divergence of $\sum a_n$.

EXAMPLE 1 Test the series $\sum_{n=1}^{\infty} (-1)^n \frac{n^3}{3^n}$ for absolute convergence.

Solution:

EXAMPLE 2 Test the convergence of the series $\sum_{n=1}^{\infty} \frac{n^n}{n!}$.

Solution:

EXAMPLE 3

$$\sum_{n=1}^{\infty} \frac{1}{n}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2}$$

11.7 | Strategy for Testing Series

1. **Test for Divergence** If you can see that $\lim_{n \rightarrow \infty} a_n$ may be different from 0, then apply the Test for Divergence.
2. **p -Series** If the series is of the form $\sum 1/n^p$, then it is a p -series, which we know to be convergent if $p > 1$ and divergent if $p \leq 1$.
3. **Geometric Series** If the series has the form $\sum ar^{n-1}$ or $\sum ar^n$, then it is a geometric series, which converges if $|r| < 1$ and diverges if $|r| \geq 1$. Some preliminary algebraic manipulation may be required to bring the series into this form.
4. **Comparison Tests** If the series has a form that is similar to a p -series or a geometric series, then one of the comparison tests should be considered. In particular, if a_n is a rational function or an algebraic function of n (involving roots of polynomials), then the series should be compared with a p -series. Notice that most of the series in Exercises 11.4 have this form. (The value of p should be chosen as in Section 11.4 by keeping only the highest powers of n in the numerator and denominator.) The comparison tests apply only to series with positive terms, but if $\sum a_n$ has some negative terms, then we can apply a comparison test to $\sum |a_n|$ and test for absolute convergence.

- 5. Alternating Series Test** If the series is of the form $\sum (-1)^{n-1} b_n$ or $\sum (-1)^n b_n$, then the Alternating Series Test is an obvious possibility. Note that if $\sum b_n$ converges, then the given series is absolutely convergent and therefore convergent.
- 6. Ratio Test** Series that involve factorials or other products (including a constant raised to the n th power) are often conveniently tested using the Ratio Test. Bear in mind that $|a_{n+1}/a_n| \rightarrow 1$ as $n \rightarrow \infty$ for all p -series and therefore all rational or algebraic functions of n . Thus the Ratio Test should not be used for such series.
- 7. Root Test** If a_n is of the form $(b_n)^n$, then the Root Test may be useful.
- 8. Integral Test** If $a_n = f(n)$, where $\int_1^\infty f(x) dx$ is easily evaluated, then the Integral Test is effective (assuming the hypotheses of this test are satisfied).

EXAMPLE 1 $\sum_{n=1}^{\infty} \frac{n-1}{2n+1}$

EXAMPLE 2 $\sum_{n=1}^{\infty} \frac{\sqrt{n^3+1}}{3n^3+4n^2+2}$

EXAMPLE 3 $\sum_{n=1}^{\infty} ne^{-n^2}$

EXAMPLE 4 $\sum_{n=1}^{\infty} (-1)^n \frac{n^2}{n^4+1}$

EXAMPLE 5 $\sum_{k=1}^{\infty} \frac{2^k}{k!}$

EXAMPLE 6 $\sum_{n=1}^{\infty} \frac{1}{2+3^n}$

